
ALGEBRA-SUMMER

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0.1 Intro

Theorem 0.1.1 (Division Algoritm). Suppose $a, b \in \mathbb{Z}$ s.t. $b > 0$. Then there are unique $q, r \in \mathbb{Z}$ s.t. $a = bq + r$ and $0 \leq r < b$

Proof. Let $a, b \in \mathbb{Z}$ s.t. $b > 0$. Consider $A = \{a - bx \mid x \in \mathbb{Z}\} \cap \mathbb{N}$

By the Well Ordering Principle since A has all non-negative terms it has a minimum lets call this r and call the associated $x \in \mathbb{Z}$, q .

Thus, $r = a - bq \implies a = bq + r$.

For contradiction, assume $r \geq b$.

$\implies r = b + m$ s.t. $m \geq 0$

$\implies b + m = a - bq \implies m = a - b(q + 1) \in A \implies m < r = \text{minimum} \implies \Leftarrow$ Thus $0 \leq r < b$

Fix $q_1, q_2, r_1, r_2 \in \mathbb{Z}$ s.t. $a = bq_1 + r_1 = bq_2 + r_2$ and $0 \leq r_1, r_2 < b$. WLOG let $r_1 \geq r_2$

$\implies bq_1 + r_1 = bq_2 + r_2 \implies r_1 - r_2 = b(q_2 - q_1) \implies r_1 - r_2$ is a multiple of b

Since $0 \leq r_1, r_2 < b \implies 0 \leq r_1 - r_2 < b$

Since $r_1 - r_2$ is strictly less than b the only multiple of b it could be is 0 thus $r_1 = r_2$

$\implies q_1 = q_2$ □

Lemma 0.1.2. (Bezout's Lemma) Suppose $a, b \in \mathbb{Z} \setminus \{0\}$. Then there are $x, y \in \mathbb{Z}$ s.t. $ax + by = \gcd(a, b)$.

Proof. Let $a, b \in \mathbb{Z} \setminus \{0\}$. Consider $A = \{ax + by \mid x, y \in \mathbb{Z}\} \cap \mathbb{N}$

By the Well Ordering Principle since A has all non-negative terms it has a minimum lets call this d and call the associated $x, y \in \mathbb{Z}$, u and v .

By Theorem 0.1.1 $\exists q, r \in \mathbb{Z}$ s.t. $a = dq + r$ and $0 \leq r < d$

Case 1: $r = 0 \implies d \mid a$

Case 2: $0 < r < d \implies r = a - dq \implies r = a - (au + bv)q = a(1 - uq) + b(-vq) \in A$

$\implies r < d = \text{minimum} \implies \Leftarrow$

Thus $d \mid a$ and similarly $d \mid b$

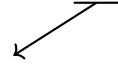
Let $c \in \mathbb{N}$ s.t. $c \mid a$ and $c \mid b$

$\implies a = cm$ and $b = cn$ for $m, n \in \mathbb{Z}$

$d = au + bv = c(mu) + c(mv) = c(mu + mv) \implies c \mid d \implies d = \gcd(a, b)$ □

Algorithm 0.1.3. (Euclidean Algorithm)Find $\gcd(272, 571)$ and $x, y \in \mathbb{Z}$ s.t. $272x + 571y = \gcd(272, 571)$

$$571 = 272 \cdot 2 + 27 \quad (1)$$



$$272 = 27 \cdot 10 + 2 \quad (2)$$



$$27 = 2 \cdot 13 + 1 \quad (3)$$

Last term before remainder is zero is gcd

$$\gcd(272, 571) = 1$$

$$2 = 272 - 27 \cdot 10$$

Use equation (2)

$$= 272(21) - 571(10)$$

$$1 = (571 - 272 \cdot 2) - 13(272(21) - 571)$$

Use equation (3) with (2)

$$= 571(131) + 272(-275)$$

1 Groups

1.1 Intro

Definition 1.1.1. A group is a set G together with a binary operation $*$: $G \times G \rightarrow G$ s.t.

- **(Closure)** For all $a, b \in G$ we have $a * b \in G$
- **(Identity)** There is $e \in G$ s.t. for all $a \in G$ $a * e = e * a = a$
- **(Inverses)** For all $a \in G$ there is an $a^{-1} \in G$ s.t. $a * a^{-1} = a^{-1} * a = e$
- **(Associativity)** For all $a, b, c \in G$ we have $a * (b * c) = (a * b) * c$

Definition 1.1.2. Given a group G and $g \in G$ we say the order of g is $n \in \mathbb{N}$ if n is the minimum value s.t. $g^n = g * g * \dots * g = e$
Write as: $\text{ord}(g) = |g| = n$

1.1.1 Exercises

Example 1.1.3. Let $G = \mathbb{R} \setminus \{-1\}$ and define $a * b = a + b + ab$. Show $(G, *)$ is an Abelian group

- (Abelian) For all $a, b \in G$ $a * b = a + b + ab = b + a + ba = b * a$
- (Closure) $\checkmark$
- (Identity) For all $a \in G$ $a * 0 = 0 * a = 0$
- (Inverses) For all $a \in G$ let $b = \frac{-a}{a+1}$.

$$\text{Thus, } a * b = b * a = a + b + ab = a + b(a+1) = \frac{-a}{a+1}(a+1) = a - a = 0$$

- (Associativity) For all $a, b, c \in G$

$$a * (b * c) = a * (b + c + bc) = a + b + c + bc + ab + ac + abc = (a + b + ab) * c = (a * b) * c$$

Therefore, G is an Abelian group

Example 1.1.4. Find all $x \in \mathbb{Z}$ s.t. $5x \equiv 4 \pmod{9}$

$$5x \equiv 4 \pmod{9}$$

$$x \equiv 8 \pmod{9}$$

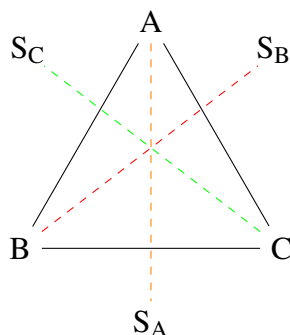
$$x = 9n + 8$$

$$2 * 5 = 10 \equiv 1 \pmod{9}$$

$$n \in \mathbb{Z}$$

1.2 Dihedral Groups

1.2.1 Rigid Motions of an Equilateral Triangle



Rotations

$r = 120^\circ$ rotation

$r^2 = 240^\circ$ rotation

$r^3 = 360^\circ$ rotation $= e = 0^\circ$ rotation

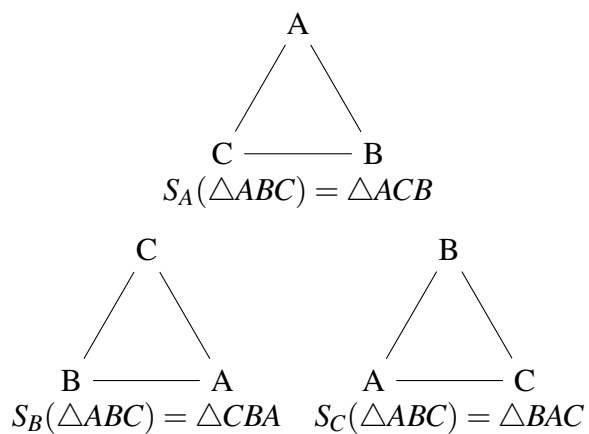
Reflections

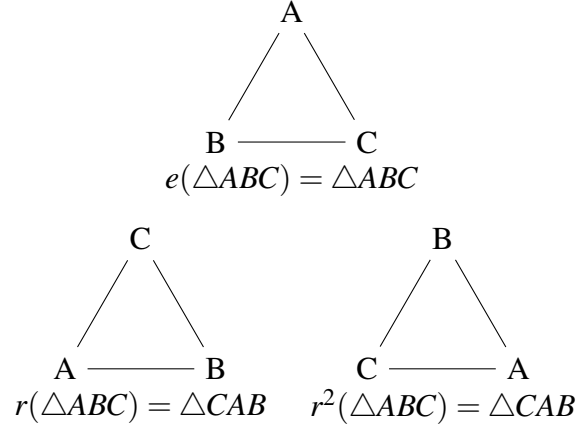
$S_A : \triangle ABC \rightarrow \triangle ACB$

$S_B : \triangle ABC \rightarrow \triangle CBA$

$S_C : \triangle ABC \rightarrow \triangle BAC$

Transformation Diagrams





Operations

$$\begin{aligned}
 r \circ S_A(\triangle ABC) &= r(\triangle ACB) = \triangle BAC = S_C(\triangle ABC) & r \circ S_A &= S_C \\
 r^2 \circ S_A(\triangle ABC) &= r^2(\triangle ACB) = \triangle CBA = S_B(\triangle ABC) & r^2 \circ S_A &= S_B
 \end{aligned}$$

Now we only need S_A so lets call this S

Thus we only need S, r, r^2, e

Also $S^2 = r^3 = e$

How S, r, r^2 Commute

$$\begin{aligned}
 r \circ S(\triangle ABC) &= r(\triangle ACB) = \triangle BAC = S \circ r^2 \\
 r^2 \circ S(\triangle ABC) &= r^2(\triangle ACB) = \triangle CBA = S \circ r
 \end{aligned}$$

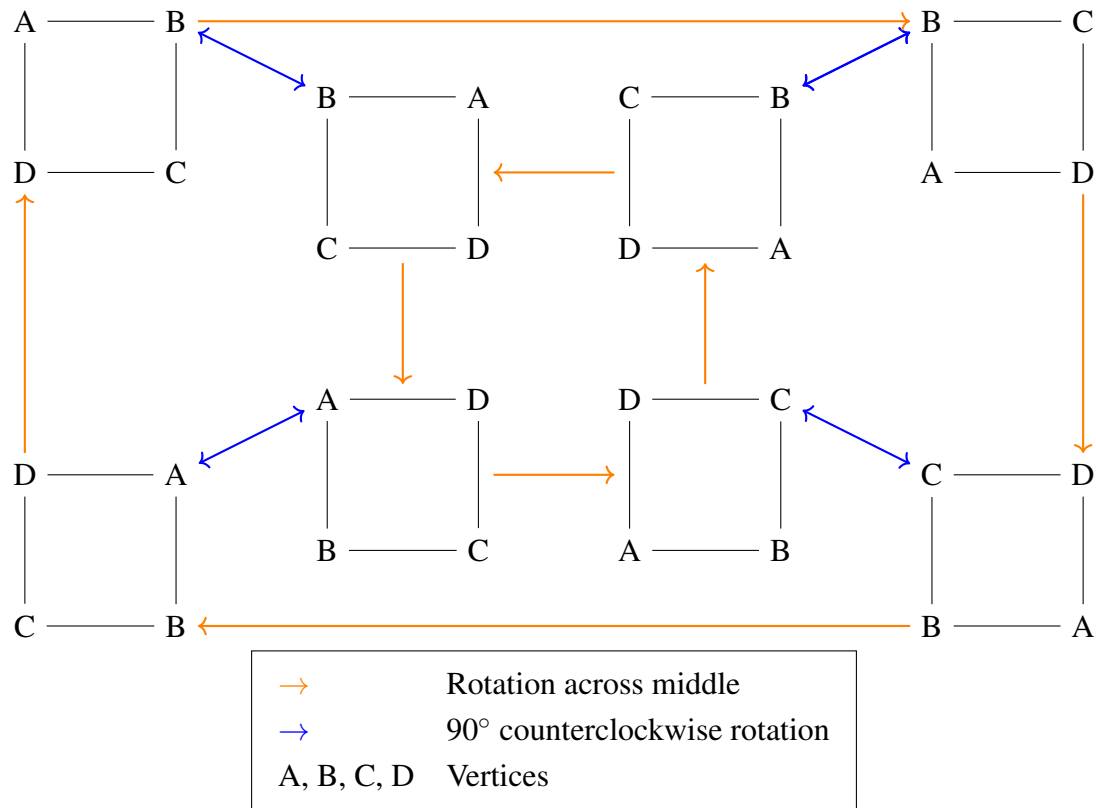
Definition 1.2.1. The group of symmetries of an equilateral triangle is $D_3 = \{e, r, r^2, S, Sr, Sr^2\}$, where $r^3 = S^2 = e$, $rS = Sr^2$

Dihedral group of order $2 \cdot 3 = 6$

Compact Notation $D_3 = \langle r, s^1 \mid r^3 = S^2 = e, rS = Sr^2 \rangle$

¹The generators of D_3 meaning you can create all other operations from them

1.2.2 Symmetry Group for a Square



Rotations

$r = 90^\circ$ rotation
 $r^2 = 180^\circ$ rotation
 $r^3 = 270^\circ$ rotation
 $r^4 = 360^\circ$ rotation = $e = 0^\circ$ rotation

Reflections

$$s : \square ABCD \rightarrow \square BCDA$$

How does s, r Commute

From the diagram we see that $sr = r^3s \implies rs = sr^3$

Definition 1.2.2. The group of symmetries of a square is $D_4 = \{e, r, r^2, r^3, s, sr, sr^2, sr^3\}$

Dihedral group of order $2 \cdot 4 = 8$

$$D_4 = \langle s, r \mid r^4 = s^2 = e, sr = r^3s \rangle$$

1.2.3 Symmetry Group of regular n-gon

Rotation: $r = \frac{360^\circ}{n} \circlearrowleft (e, r, r^2, \dots, r^{n-1})$

Reflection: $s = \text{reflection across middle } (s, sr, sr^2, \dots, sr^{n-1})$

Claim: $rs = sr^{n-1}$.

Proof. We only need 3 points since our shape is defined by the order of those 3 points

$$r \circ s(n, 1, 2) = r(2, 1, n) = (1, n, n-1)$$

$$s \circ r^{n-1}(n, 1, 2) = s(n-1, n, 1) = (1, n, n-1)$$

Thus $rs = sr^{n-1}$ □

Definition 1.2.3. The dihedral group D_n is the symmetry group of a regular n-gon.

$$D_n = \langle s, r \mid r^n = s^2 = e, rs = sr^{n-1} \rangle$$

Claim: For all $1 \leq m \leq n-1$ we have $r^m s = sr^{n-m}$.

Proof. Base Case: $m = 1 \implies rs = sr^{n-1}$ ✓

Inductive Step: Assume for $k \geq 1$ the statement is true

Consider:

$$r^{k+1}s = rr^k s$$

$$= rsr^{n-k}$$

$$= sr^{n-1}r^{n-k}$$

$$= sr^n r^{n-(k+1)}$$

$$= sr^{n-(k+1)}$$

Apply inductive hypothesis on $r^k s$

□

1.2.4 Exercises

Example 1.2.4. Simplify

- $(sr^5)r(sr^3)$ in D_4

$$= (sr)r(rs)$$

$$= e$$

- $(sr^4)(sr^2)$ in D_6

$$= (r^2s)(sr^2) \quad \text{Use } r^m s = sr^{n-m}$$

$$= r^4$$

- $r^3(sr)$ in D_4

$$= r^3(r^3s)$$

$$= r^2s$$

Example 1.2.5. The Cayley table of the D_3 can be written

	e	r	r^2	s	rs	r^2s
e	e	r	r^2	s	rs	r^2s
r	r	r^2	e	rs	r^2s	s
r^2	r^2	e	r	r^2s	s	rs
s	s	r^2s	rs	e	r^2	r
rs	rs	s	r^2s	r	e	r^2
r^2s	r^2s	rs	s	r^2	r	e

1.3 Symmetric Groups

Definition 1.3.1. Given a set X , a permutation of X is a bijection $X \rightarrow X$
 $S_X = \{ \sigma : X \rightarrow X \mid \sigma \text{ is a bijection} \}$

Lemma 1.3.2. For all sets $X \neq \emptyset$, S_X is a group

Proof.

- Closure: Suppose $\sigma, \mu \in S_X$
 Consider $\sigma \circ \mu$ since it is a composition of bijections it is also a bijection.
 Also functions go from $X \rightarrow X$ thus $\sigma \circ \mu$ is the same.
 Therefore, the composition is a bijection that is mapped from $X \rightarrow X$ thus $\sigma \circ \mu \in S_X$
- Identity: Consider $e : X \rightarrow X$ s.t. $e(x) \mapsto x$
 Thus $e \in S_X$ and for all $\sigma \in S_X$ $\sigma \circ e(x) = \sigma(x)$
 Thus $\sigma \circ e = \sigma$. Also, it follows that $e \circ \sigma = \sigma$
- Inverses: Since S_X has only bijective functions all bijective functions have inverses.
- Associativity: We have associativity by the fact that function composition has associativity.

□

Definition 1.3.3. The permutation group on n -objects S_n is the S_X with $X = [n]$
 Observe $|S_n| = n!$

Example 1.3.4. $\sigma, \mu \in S_4$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \mu = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix}$$

$$\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \mu^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix}$$

$$\sigma\mu = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \mu\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix}$$

1.3.1 Cycle Notation

Example 1.3.5. $\sigma, \mu \in S_5$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 2 & 5 & 1 \end{pmatrix} \mu = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 2 & 1 & 3 \end{pmatrix}$$

$$\sigma = (13245) \mu = (14)(253)$$

Example Calculations:

$$\sigma\mu = (13245)(14)(253)$$

$$= (152)(34)$$

$$\mu\sigma = (14)(253)(13245)$$

$$= (12)(354)$$

$$\mu^2 = (14)(253)(14)(253)$$

$$= (14)^2(253)^2 = (235)$$

Lemma 1.3.6. *Disjoint cycles are commutative*

Proof. Let α, β be disjoint cycles of $[n]$

Let $\alpha = (a_1, a_2, \dots, a_k)$, and $\beta = (b_1, b_2, \dots, b_j)$

Let $x \in [n]$

Consider: $\alpha\beta(x), \beta\alpha(x)$

If $x \in \alpha$ then $x \notin \beta \implies \beta(x) = x \implies \alpha\beta(x) = \alpha(x) = a_i$

$\implies \beta\alpha(x) = \beta(a_i) = a_i$ since $a_i \notin \beta$

Use similar reasoning for if $x \in \beta$

$\implies \alpha\beta = \beta\alpha$

□

Lemma 1.3.7. *Every $\sigma \in S_n$ can be written as disjoint cycles*

Lemma 1.3.8. *The order of an m -cycle is m*

Lemma 1.3.9. Suppose G is a group and $\text{ord}(g) = n$. Then $g^m = e \iff n \mid m$

Proof. Let G be a group and $\text{ord}(g) = n$

($\Leftarrow$)

Assume $n \mid m$

$$\implies m = nk, k \in \mathbb{Z}$$

$$g^m = g^{nk} = (g^n)^k = (e)^k = e$$

($\Rightarrow$)

Assume $g^m = e$ and use the division algorithm (0.1.1) to write $m = nq + r$ s.t. $0 \leq r < n$

$$e = g^m = g^{nq+r} = (g^n)^q g^r = g^r \implies r = 0 \implies n \mid m$$

□

Theorem 1.3.10. If $\sigma_1, \sigma_2, \dots, \sigma_k$ are disjoint cycles of length $a_1, a_2, \dots, a_k$ and $\sigma = \sigma_1 \sigma_2 \dots \sigma_k$. Then $\text{ord}(\sigma) = \text{lcm}(a_1, a_2, \dots, a_k)$.

Proof. Let $m = \text{ord}(\sigma)$, $l = \text{lcm}(a_1, a_2, \dots, a_k)$

Since l is a lcm it is also a multiple so $l = a_j q_j$ for $1 \leq j \leq k$ where $q_j \in \mathbb{Z}$

$$\begin{aligned} \sigma^l &= \sigma_1^l \sigma_2^l \dots \sigma_k^l \\ &= (\sigma_1^{a_1})^{q_1} \dots (\sigma_k^{a_k})^{q_k} \\ &= (1) \end{aligned}$$

We can do this because of Lemma 1.3.6

Apply Lemma 1.3.8

$$\implies m \mid l$$

Also, $(1) = \sigma^m = \sigma_1^m \dots \sigma_k^m$ since they are disjoint none of them are inverses of each other

$$\implies \sigma_1^m = \dots = \sigma_k^m = (1)$$

$$(\text{Lemma 2.1.9}) \implies a_1 \mid m \dots a_k \mid m \implies l \mid m$$

Thus, since $l \mid m$ and $m \mid l$ $m = l$

□

Example 1.3.11.

- Find an element of order 12 in S_7
 $(1234)(567) \in S_7$ by Theorem 1.3.10 $\implies$ the order is $3 \cdot 4 = 12 = \text{lcm}(3,4)$
- Possible orders of elements in S_5
 - 1: (1) , $5=1+1+1+1+1$
 - 2: (12) , $5=2+1+1+1+1$ or $(12)(34)$, $5=2+2+1$
 - 3: (123) , $5=3+1+1$
 - 4: (1234) , $5=4+1$
 - 5: (12345) , $5=5$
 - 6: $(12)(345)$, $5=2+3$

1.3.2 Exercises

Example 1.3.12. Write the following as the product of disjoint cycles and find the order.

- $(1358)(378)(125) \in S_8$
 (128537) order=6
- $(1264)^{-1}(364)^2 \in S_6$
 $= (1462)(346)$
 $= (142)(36)$ order=6

Example 1.3.13. Find all Possible orders of elements in S_7

- 1: (1) , $7=1+1+1+1+1+1+1$
- 2: (12) , $7=2+1+1+1+1+1$ or $(12)(34)$, $7=2+2+1+1+1$ or $(12)(34)(56)$, $7=2+2+2+1$
- 3: (123) , $7=3+1+1+1+1$ or $(123)(456)$, $7=3+3+1$
- 4: (1234) , $7=4+1+1+1$ or $(1234)(56)$, $7=4+2+1+1$
- 5: (12345) , $7=5+1+1$
- 6: (123456) , $7=6+1$ or $(123)(45)$, $7=3+2+1+1$, or $(123)(45)(67)$, $7=3+2+2+1$
- 7: (1234567) , $7=7$
- 10: $(12345)(67)$, $7=5+2$
- 12: $(1234)(567)$, $7=4+3$

Example 1.3.14. What is the largest possible order of an element in S_{10} ?

cycle type	order
(5,1,1,1,1)	5
(5,2,1,1,1)	10
(5,2,2,1)	10
(5,3,1,1)	15
(5,3,2)	30
(5,4,1)	20
(5,5)	5
(6,1,1,1,1)	6
(6,2,1,1)	6
(6,3,1)	6
(6,4)	12
(7,1,1,1)	7
(7,2,1)	14
(7,3)	21
(8,1,1)	8
(8,2)	8
(9,1)	9
(10)	10

Max=30

1.4 General Results on Groups

Lemma 1.4.1. *Suppose G is a group. Then*

- (1) *The identity is unique*
- (2) *Inverses are unique*
- (3) *For every $a, b, c \in G$ there is a unique $x \in G$ s.t. $ax = b$*
- (4) *If $ab = ac$ then $b = c$, similarly if $xy = zy$ then $x = z$*
- (5) $(ab)^{-1} = b^{-1}a^{-1}$
- (6) *If $(ab)^2 = a^2b^2$ then $ab = ba$*
- (7) *If $\text{ord}(g) = 2$ for all $g \in G$ then G is abelian*

• *Proof.* (3) Let $ax = b \implies x = a^{-1}b$

Suppose $x, y \in G$ s.t. $ax = b$ and $ay = b$

$$ax = ay \implies x = y \quad \square$$

• *Proof.* (6) Assume $(ab)^2 = a^2b^2$

$$ab = (a^{-1}a)(ab)(bb^{-1}) = a^{-1}a^2b^2b^{-1} = a^{-1}(ab)^2b^{-1} = (a^{-1}a)(ba)(bb^{-1}) = ba \quad \square$$

• *Proof.* Assume $\text{ord}(g) = 2$ for all $g \in G$

$$\implies g = g^{-1} \text{ for all } g \in G$$

Let $a, b \in G$. Consider ab

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba \quad \square$$

Definition 1.4.2. Given a group G and a subset $H \subseteq G$, we say H is a subgroup $H \leq G$ if

- The identity of G is in H
- For all $h \in H, h^{-1} \in H$
- For all $a, b \in H, ab \in H$

Theorem 1.4.3. Suppose that G is a group and $\emptyset \neq H \subseteq G$
 then $H \leq G \iff$ for all $a, b \in H, ab^{-1} \in H$

Proof.

($\implies$) Assume $H \leq G$

Let $a, b \in H$

By Definition 1.4.2 $\implies b^{-1} \in H \implies ab^{-1} \in H$

($\impliedby$) Assume for all $a, b \in H, ab^{-1} \in H$

- (Identity) Consider $a \in H$

$$\implies aa^{-1} = e \in H$$

- (Inverses) $\checkmark$

- (Closure) Consider $a, b^{-1} \in H$

$$\implies ab \in H$$

□

1.4.1 Subgroup Lattice

Subgroup of Order 4

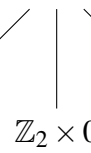
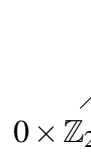
$$\mathbb{Z}_4 = \{0, 1, 2, 3\}$$



$$\begin{array}{c} | \\ 0 = \{0\} \end{array}$$

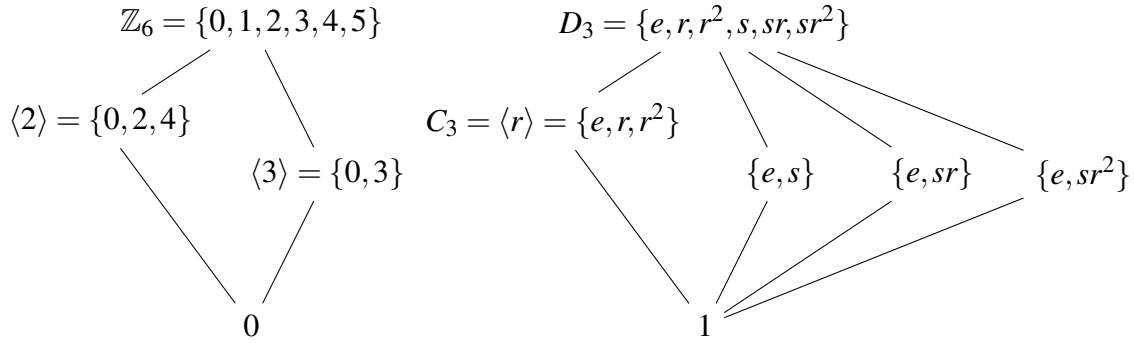
$$0 = \{0\}$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 = \{(0, 0), (1, 0), (0, 1), (1, 1)\}$$



$$0 = \{(0, 0)\}$$

Subgroup of Order 6



Definition 1.4.4. Given a group G and a subset $S \subseteq G$

- The centralizer of S in G : $C_G(S) = \{g \in G \mid gs = sg, \forall s \in S\}$
- The center of G : $(C_G(G)) = Z(G) = \{z \in G \mid gz = zg, \forall g \in G\}$

Lemma 1.4.5. $C_G(S)$ and $Z(G)$ are subgroups of G

Proof. Both contain e so they are nonempty

Let $a, b \in C_G(S)$ and $s \in S$

Consider:

$$\begin{aligned} as &= sa \\ sab^{-1} &= asb^{-1} \\ &= a(b^{-1}s)b \\ &= (ab^{-1})s \end{aligned}$$

Thus $ab^{-1} \in C_G(S)$

Thus, by Theorem 1.4.3 $C_G(S)$ is a subgroup

$\implies Z(G) \leq G$ since the center is a special case of the centralizer

□

Example 1.4.6. Centralizers in D_4

- $C_{D_4}(e) = D_4 = C_{D_4}(r^2)$ ²
- $C_{D_4}(r) = e, r, r^2, r^3 = C_{D_4}(r^3)$
- $C_{D_4}(s) = e, s, r^2, sr^2 = C_{D_4}(sr^2)$
- $C_{D_4}(sr) = e, sr, r^2, sr^3 = C_{D_4}(sr^3)$
- $Z(D_4) = e, r^2$

²Since $r^2s = sr^2$ and $r^2r = rr^2$

1.4.2 Exercises

Example 1.4.7. Prove that if G is abelian if and only if for all $a, b \in G$ we have $(ab)^{-1} = a^{-1}b^{-1}$

Proof. ($\implies$) Assume G is abelian and let $a, b \in G$

$$\implies (ab)^{-1} = b^{-1}a^{-1} = a^{-1}b^{-1}$$

($\impliedby$) Assume for all $a, b \in G$ we have $(ab)^{-1} = a^{-1}b^{-1}$

Consider: a^{-1} and b^{-1}

$$ba = (a^{-1}b^{-1})^{-1} = ab$$

□

Example 1.4.8. Suppose G is a group of order 6 containing 7 and 19 where the operation is multiplication modulo 36. List all the elements of G .

- $7 \times 7 = 49 = 13$
- $19 \times 19 = 361 = 1$
- $7 \times 19 = 133 = 25$
- $13 \times 19 = 247 = 31$

$$G = \{1, 7, 13, 19, 25, 31\}$$

Example 1.4.9. Suppose that G is a group such that for all $a, b, c, d \in G$ such that if

$$axb = cxd$$

then $ab = cd$. Show that G is abelian

Proof. Consider: $a, b \in G$

$$aa^{-1}b = ba^{-1}a \implies ab = ba$$

□

1.5 Cyclic Groups

Definition 1.5.1. Given a group G and an element $g \in G$, the cyclic subgroup generated by g is $\langle g \rangle = \{g^m \mid m \in \mathbb{Z}\} = \{\dots, g^{-2}, g^{-1}, e, g, g^2, \dots\}$

We say G is a cyclic group if there is a $g \in G$ s.t. $G = \langle g \rangle$

$$\text{ord}(g) = |\langle g \rangle|$$

Cyclic subgroups of $\mathbb{Z}$

Example 1.5.2.

- $\langle 2 \rangle = \{2m \mid m \in \mathbb{Z}\} = \{\dots, -4, -2, 0, 2, 4, \dots\} = 2\mathbb{Z}$
- $\langle n \rangle = \{nm \mid m \in \mathbb{Z}\} = \{\dots, -2n, -n, 0, n, 2n, \dots\} = n\mathbb{Z}$

Cyclic subgroups of $\mathbb{Q}^\times$

Example 1.5.3.

- $\langle -1 \rangle = \{-1, 1\}$
- $\langle 2 \rangle = \{2^m \mid m \in \mathbb{Z}\}$

Cyclic subgroups of $\mathbb{C}^\times$

Example 1.5.4.

$$\langle i \rangle = \{i, -1, -i, 1\}$$

Cyclic subgroups of $\mathbb{Z}_{12}$

Example 1.5.5.

- $\mathbb{Z}_{12} = \langle 1 \rangle = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$
- $\langle 2 \rangle = \{0, 2, 4, 6, 8, 10\}$
- $\langle 4 \rangle = \{0, 4, 8\}$
- $\langle 3 \rangle = \{0, 3, 6, 9\}$

Cyclic subgroups of $U_9 = \{1, 2, 4, 5, 7, 8\}$

Example 1.5.6.

- $\langle 1 \rangle = \{1\}$
- $\langle 8 \rangle = \{8, 64\} = \{1, 8\}$
- $\langle 4 \rangle = \{4, 16, 64\} = \{1, 4, 7\}$
- $\langle 2 \rangle = \{2, 4, 8, 16, 32, 64\} = \{1, 2, 4, 5, 7, 8\} = U_9$

Cyclic subgroups of D_n

Example 1.5.7.

- $\langle r \rangle = \{e, r, r^2, \dots, r^{n-1}\} = Z_n^3 = C_n^4$
- $\langle s \rangle = \{e, s\}$
- $\langle sr \rangle = \{e, sr\}$

Cyclic subgroups of S_5

Example 1.5.8.

$$\langle (12)(345) \rangle^5 = \{(1), (12)(345), (354), (12), (345), (12)(354)\}$$

Theorem 1.5.9. Suppose $g \in G$ with $\text{ord}(g) = n$ and let $m \in \mathbb{N}$ then, $\langle g^m \rangle = \langle g^{\gcd(m,n)} \rangle$

Proof. Set $d = \gcd(m, n)$.

Take $x \in \langle g^m \rangle \implies x = (g^m)^k = g^{mk}$ for some $k \in \mathbb{Z}$

Also, $m = dq$ for some $q \in \mathbb{Z}$ since d is a divisor of m

$x = g^{dqk} = (g^d)^{qk} \in \langle g^d \rangle$ so $\langle g \rangle \subseteq \langle g^d \rangle$

By Lemma 0.1.2 there exists $r, s \in \mathbb{Z}$ s.t. $mr + ns = d$

Take $x \in \langle g^d \rangle \implies x = g^{dk}$ for some $k \in \mathbb{Z}$

$g^{dk} = g^{k(mr+ns)} = g^{mkr}(g^n)^{ks} = g^{mkr} \in \langle g^m \rangle$ so $\langle g^d \rangle \subseteq \langle g^m \rangle$

Thus $\langle g^m \rangle = \langle g^d \rangle$

□

³not $\mathbb{Z}$

⁴Multiplicative cyclic group of order n

⁵By Theorem 1.3.10 the $\text{lcm}(2,3)=6=\text{order}$ and thus this cyclic group will have 6 elements

Theorem 1.5.10 (Fundamental Theorem of Cyclic Groups).

- I. Every subgroup of a cyclic group is cyclic
- II. If G is a cyclic group of order n and $d \mid n$ then there is a unique subgroup $H \leq G$ with $|H| = d$

I. *Proof.* Suppose $G = \langle g \rangle$ and $H \leq G$

Case 1: $H = \{e\} = \langle e \rangle \checkmark$

Case 2: $H \neq \{e\}$. Thus, $H = \{\dots, g^{-m_2}, g^{-m_1}, e, g^{m_1}, g^{m_2}, \dots\}$ s.t. $m_i \in \mathbb{N}$

By the Well Ordering Principle, there exists a minimum m_i call this m

Since $g^m \in H \implies (g^m)^n \in H \ \forall n \in \mathbb{Z} \implies \langle g^m \rangle \subseteq H$

Take $h \in H$, write $h = g^k$ for $k \in \mathbb{Z}$

By the division algorithm (Thm. 0.1.1), there exists a $q, r \in \mathbb{Z}$ s.t.

$k = mq + r$ with $0 \leq r < m$

$g^r = (g^k)(g^{-mq})$ thus since $g^k \in H$ and $g^{-mq} \in \langle g^m \rangle \subseteq H \implies g^r \in H$

Since $0 \leq r < m$ and m is the minimum $\implies r = 0$

Thus $h = (g^m)^q \in \langle g^m \rangle \implies H \subseteq \langle g^m \rangle$

$\implies H = \langle g^m \rangle \implies H$ is cyclic □

II. *Proof.* Assume G is a cyclic group of order n and $d \mid n$

Let $G = \langle g \rangle$

Existence: Consider $\langle g^{\frac{n}{d}} \rangle$ this has d elements so $\checkmark$

Uniqueness: Let $H \leq G$, $|H| = d$ by Theorem 1.5.10.I

$\implies H = \langle g^m \rangle$ by Theorem 1.5.9 $\implies H = \langle g^{\gcd(m,n)} \rangle$

Thus $|H| = d = \frac{n}{\gcd(m,n)} \implies \gcd(m,n) = \frac{n}{d} \implies H = \langle g^{\frac{n}{d}} \rangle$ which unique describes a group □

Definition 1.5.11 (Euler's Totient Function).

For $n \in \mathbb{N}$, define $\varphi(n) = \left| \{1 \leq m \leq n \mid \gcd(m,n) = 1\} \right|$

Theorem 1.5.12.

Suppose G is a cyclic group of order n and $d \mid n$ then the number of elements of G of order d is $\varphi(d)$

Proof. Assume G is cyclic group of order n and $d \mid n$

Let $H \leq G$ be the unique subgroup s.t. $|H| = d \iff$ Theorem 1.5.10.II

Let $H = \langle h \rangle \iff$ Theorem 1.5.10.I

Let $x \in G$ have order d

Consider: $\langle x \rangle$ since x is order $d \implies |\langle x \rangle| = d$

Since H is unique $\implies \langle h \rangle = \langle x \rangle \implies x = h^m$ where $\gcd(m, d) = 1 \iff$ Theorem 1.5.9

The number of choices of m is $\varphi(d)$

□

1.5.1 Exercises

Example 1.5.13. Find all generators of $\mathbb{Z}_{15}$ and $\mathbb{Z}_{12}$

By Theorem 1.5.9 this implies all the numbers that are relatively prime to 12 and 15

Thus, the generators for $\mathbb{Z}_{15}$ are 1,2,4,7,8,11,13,14

Thus, the generators for $\mathbb{Z}_{12}$ are 1,5,7,11

Example 1.5.14. Show that $\mathbb{Q}_{\geq 0}^{\times}$ is not cyclic

Assume $\mathbb{Q}_{\geq 0}^{\times}$ is cyclic thus there exists a $q = \frac{a}{b} \in \mathbb{Q}_{\geq 0}^{\times}$ s.t. $\mathbb{Q}_{\geq 0}^{\times} = \langle q \rangle$

Consider: p a prime bigger than a, b

Let $\frac{1}{p} = \frac{a^m}{b^m}$

Case 1: $m \geq 1$

$\implies b^m = pa^m \implies p \mid b^m \implies \Leftarrow$ since we picked p to be bigger than a, b

Case 2: $m \leq -1$

$\implies a^m = pb^m \implies p \mid a^m \implies \Leftarrow$

1.6 Homomorphism and Isomorphism

Definition 1.6.1. Let $(G_1, *)$ and $(G_2, \circ)$ be groups. A map $\varphi : G_1 \rightarrow G_2$ is called a homomorphism if for all $a, b, c \in G_1$ we have $\varphi(a * b) = \varphi(a) \circ \varphi(b)$

If, in addition, φ is a bijection we say φ is an isomorphism and G_1 and G_2 are isomorphic

$$G_1 \cong G_2$$

Theorem 1.6.2. Suppose $G = \langle g \rangle$ is a cyclic group.

If $|G| = \infty$ then $G \cong \mathbb{Z}$

If $|G| = n$ then $G \cong \mathbb{Z}_n$

Proof. Assume $|G| = \infty$ Define:

$$\begin{aligned} \varphi : G &\rightarrow \mathbb{Z} \\ g^m &\mapsto m \end{aligned}$$

$$\begin{aligned} \varphi^{-1} : \mathbb{Z} &\rightarrow G \\ m &\mapsto g^m \end{aligned}$$

Let $g^a, g^b \in G$ and $c, d \in \mathbb{Z}$

Consider: $\varphi(g^a g^b) = \varphi(g^{a+b}) = a + b = \varphi(g^a) + \varphi(g^b)$

Consider: $\varphi^{-1}(c + d) = g^{c+d} = g^c g^d = \varphi^{-1}(c) + \varphi^{-1}(d)$

Consider: $\varphi \circ \varphi^{-1}(c) = \varphi(g^c) = c$

$\varphi^{-1} \circ \varphi(g^a) = \varphi^{-1}(a) = g^a$

Thus, φ is a homomorphism and bijective

$$\implies G \cong \mathbb{Z}$$

Assume $|G| = n$

$$\begin{aligned} \varphi : G &\rightarrow \mathbb{Z}_n \\ g^m &\mapsto [m] \end{aligned}$$

$$\begin{aligned} \varphi^{-1} : \mathbb{Z}_n &\rightarrow G \\ [m] &\mapsto g^m \end{aligned}$$

homomorphism: ✓

Consider: $\varphi \circ \varphi^{-1}([c]) = \varphi(g^c) = [c]$

$\varphi^{-1} \circ \varphi(g^a) = \varphi^{-1}([a]) = g^a$

Thus, φ is a homomorphism and bijective

$\implies G \cong \mathbb{Z}_n$

□

Theorem 1.6.3 (Cayley's Theorem). *Every group is isomorphic to a permutation group.*

Proof. Let G be a group and by Lemma 1.3.2 $\implies S_G$ is a group

Define:

$$\varphi : G \rightarrow S_G$$

$$g \mapsto l_g$$

$$l_g : G \rightarrow G$$

$$x \mapsto gx$$

- **Injective:** Let $x, y \in G$ s.t. $l_g(x) = l_g(y)$

$$\implies gx = gy \implies x = y$$

- **Surjective:** Let $y \in G$

$$\text{Consider: } l_g(g^{-1}y) = gg^{-1}y = y$$

Thus $l_g \in S_G$

Consider: $a, b \in G$

$$\varphi(ab)(x) = l_{ab}(x) = abx = a(bx) = l_a \circ l_b(x) = \varphi(a)\varphi(b)(x)$$

Injective: $a, b, x \in G$ s.t. $\varphi(a) = \varphi(b)$

$$l_a = l_b \implies l_a(x) = l_b(x) \quad \forall x \in G \implies ax = bx \implies ae = be \implies a = b$$

$$\implies G \cong \text{im}(G) \leq S_G$$

□

Theorem 1.6.4. Suppose $\varphi : G_1 \rightarrow G_2$ is a homomorphism then

I. $\varphi(e_1) = e_2$

II. $\forall n \in \mathbb{Z}, \varphi(g^n) = [\varphi(g)]^n$

III. $ab = ba \implies \varphi(a)\varphi(b) = \varphi(b)\varphi(a)$

IV. $\text{ord}(\varphi(g)) \mid \text{ord}(g)$

V. $H \leq G_1 \implies \varphi(H) \leq G_2$

VI. $K \leq G_2 \implies \varphi^{-1}(K) \leq G_1$

I. *Proof.* $e_2\varphi(e_1) = e_2\varphi(e_1e_1) = e_2\varphi(e_1)\varphi(e_1) \implies e_2 = \varphi(e_1)$ □

II. *Proof.* Base Case: $n = 0$ $\varphi(g^0) = \varphi(e_1) = e_2 = [\varphi(g)]^0$

Assume statement is true for $k \geq 0$ for $k \in \mathbb{N}$

Consider: $k + 1$

$$\varphi(g^{k+1}) = \varphi(g)\varphi(g^k) = \varphi(g)[\varphi(g)]^k = [\varphi(g)]^{k+1}$$

Note: $e_2 = \varphi(e_1) = \varphi(gg^{-1}) = \varphi(g)\varphi(g^{-1}) \implies [\varphi(g)]^{-1} = \varphi(g^{-1})$

This proves the statement for all negative integers since we just apply the induction on g then find its inverse □

III. *Proof.* $\varphi(a)\varphi(b) = \varphi(ab) = \varphi(ba) = \varphi(b)\varphi(a)$ □

IV. *Proof.* Set $n = \text{ord}(g)$

$$\implies \varphi(g)^n = \varphi(g^n) = \varphi(e_1) = e_2 \implies \text{ord}(\varphi(g)) \mid \text{ord}(g)$$
 □

V. *Proof.* Let $x, y \in \varphi(H)$ s.t. $\varphi(a) = x$ and $\varphi(b) = y$ for $a, b \in H$

$$\implies ab^{-1} \in H \implies \varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in \varphi(H)$$

Thus, by Theorem 1.4.3 $\varphi(H) \leq G_2$ □

VI. *Proof.* Let $a, b \in \varphi^{-1}(K) \implies \varphi(a), \varphi(b) \in K$

$$\implies \varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in K \implies ab^{-1} \in \varphi^{-1}(K)$$

Thus, by Theorem 1.4.3 $\varphi^{-1}(K) \leq G_1$ □

Theorem 1.6.5. *thm:isomorphismgeneral Suppose $\varphi : G_1 \rightarrow G_2$ is an isomorphism then φ^{-1} is an isomorphism*

II. G_1 is abelian $\iff G_2$ is abelian

III. G_1 is cyclic $\iff G_2$ is cyclic

IV. $\text{ord}(g) = \text{ord}(\varphi(g))$

I. $\checkmark$

II. *Proof.* Suppose G_1 is abelian and $x, y \in G_2$

$$xy = \varphi(a)\varphi(b) = \varphi(ab) = \varphi(ba) = \varphi(b)\varphi(a) = yx$$

Do same thing for other direction □

III. *Proof.* Suppose G_1 is cyclic and $\langle g \rangle = G_1$

$$\text{Take } x \in G_2 \implies \varphi(a) = x$$

$$a = g^n \implies x = \varphi(g^n) = \varphi(g)^n \implies x \in \langle \varphi(g) \rangle \implies G_2 \subseteq \langle \varphi(g) \rangle$$

$$\text{Take } x \in \langle \varphi(g) \rangle \implies x = \varphi(g)^n$$

$$\implies x = \varphi(g^n) \implies x \in G_2 \implies \langle \varphi(g) \rangle \subseteq G_2$$

$$\implies G_2 = \langle \varphi(g) \rangle \implies G_2 \text{ is cyclic}$$

Do same thing for other direction □

IV. *Proof.* By Theorem 1.6.4.IV $\text{ord}(g) \mid \text{ord}(\varphi(g))$ and $\text{ord}(\varphi(g)) \mid \text{ord}(g) \implies \text{ord}(g) = \text{ord}(\varphi(g))$

□

1.6.1 Exercises

Example 1.6.6. Describe all homomorphisms $\varphi : \mathbb{Z}_{12} \rightarrow \mathbb{Z}_{20}$

$$\text{ord}(1) = 12 \text{ in } \mathbb{Z}_{12} \implies \text{ord}(\varphi(1)) \mid 12 \implies \text{ord}(\varphi(1)) \in \{1, 2, 3, 4, 6, 12\}$$

$$\text{ord}(m) = \frac{n}{\gcd(m,n)} \text{ in } \mathbb{Z}_n$$

$$\bullet \text{ ord}(\varphi(1)) = 1 \implies \varphi(1) = 0 \implies \varphi(n) = 0$$

$$\bullet \text{ ord}(\varphi(1)) = 2 \implies \varphi(1) = 10$$

$$\text{this is the only element with a gcd of 10 with 20} \implies \varphi(n) = 10n$$

$$\bullet \text{ ord}(\varphi(1)) = 3, 6, 12 \text{ there is no element with these orders since these are not divisors of 10}$$

$$\bullet \text{ ord}(\varphi(1)) = 4 \implies \varphi(1) = 5, 15 \implies \varphi(n) = 5n, 15n$$

Example 1.6.7. Show that $\mathbb{R} \not\cong \mathbb{R}^\times$

Assume $\mathbb{R} \cong \mathbb{R}^\times$ via $\varphi : \mathbb{R}^\times \rightarrow \mathbb{R}$

Take $x \in \mathbb{R}^\times$ s.t. $\varphi(x) = -1$

Consider $\varphi(\frac{x}{2})^2 = \varphi(x) = -1 \implies \varphi(\frac{x}{2}) \in \mathbb{R}$ with $\varphi(\frac{x}{2})^2 = -1 \Rightarrow \Leftarrow$

1.7 Cosets

Definition 1.7.1. Suppose G is a group $H \leq G$, and $g \in G$.

- The left coset of H with representative g is

$$gH = \{gh \mid h \in H\}$$

- The right coset of H with representative g is

$$Hg = \{hg \mid h \in H\}$$

Example 1.7.2. $G = S_3, H = \{(1), (12)\}$

$$(13)H = \{(13), (123)\} = (123)H$$

$$(23)H = \{(23), (132)\} = (132)H$$

$$H(13) = \{(13), (132)\} = H(132)$$

$$H(23) = \{(23), (123)\} = H(123)$$

Theorem 1.7.3. Suppose that G is a group, $H \leq G$, $x, y \in G$. The following are equivalent

I. $xH = yH$

II. $Hx^{-1} = Hy^{-1}$

III. $xH \subseteq yH$

IV. $y \in xH$

V. $x^{-1}y \in H$

Proof. (I. $\implies$ II.)

Assume I. Let $g \in Hx^{-1} \implies g = hx^{-1}$ for $h \in H \implies g^{-1} = xh^{-1} \in xH = yH$

$\implies g^{-1} = yj$ for $j \in H \implies g = j^{-1}y^{-1} \in Hy^{-1} \implies Hx^{-1} \subseteq Hy^{-1}$

Do the same thing for other direction thus $Hx^{-1} = Hy^{-1}$

(II. $\implies$ I.)

Assume II. Let $g \in xH \implies g = xh$ for $h \in H \implies g^{-1} = h^{-1}x^{-1} \in Hx^{-1} = Hy^{-1}$

$\implies g^{-1} = jy^{-1}$ for $j \in H \implies g = yj \in Hy \implies xH$

(III. $\implies$ IV.)

Assume III. Let $g \in xH \subseteq yH \implies g = xh = yj$ for $h, j \in H \implies y = xhj^{-1} \implies y \in xH$

(IV. $\implies$ V.)

Assume IV. $y \in xH \implies y = xh$ for $h \in H \implies x^{-1}y = h \in H$

($V \implies I.$)

Assume V. Let $g \in xH \implies g = xh$ for $h \in H \implies g = y(y^{-1}xh) \in yH$ since $y^{-1}x \in H$

Do the same thing for the other direction this $xH = yH$ □

Theorem 1.7.4. Suppose $H \leq G$. The cosets of H partition G . that is, they are disjoint and they union to G .

Proof.

Disjoint: Suppose $x, y \in G$ s.t. $xH \cap yH \neq \emptyset$

Take $g \in xH \cap yH \implies g = xh = yj$ $h, j \in H \implies x^{-1}y = hj^{-1} \in H$

Thus, by Theorem 1.7.3 $xH = yH$

Union: $\bigcup_{x \in G} xH \subseteq G \iff xH \subseteq G$

Take $g \in G$ by Theorem 1.7.3 $\implies g \in gH \implies g \in \bigcup_{x \in G} xH \implies G \subseteq \bigcup_{x \in G} xH$

$\implies G = \bigcup_{x \in G} xH$ □

Definition 1.7.5. Let G be any group with $H \leq G$.

Define the index of H in G to be the number of left(right) cosets of H in G , denote

$$[G : H]$$

Theorem 1.7.6. The number of left cosets is equal to the number of right cosets

Proof. Let $\mathfrak{L}_H = \{gH \mid g \in G\}$ and $\mathfrak{R}_H = \{Hg \mid g \in G\}$

Define:

$f: \mathfrak{L}_H \rightarrow \mathfrak{R}_H$

$gH \mapsto Hg^{-1}$

Well Defined: Suppose $xH = yH$, $f(xH) = Hx^{-1} = Hy^{-1} = f(yH)$

Injective: Suppose $f(xH) = f(yH) \implies Hx^{-1} = Hy^{-1} \implies xH = yH$

Surjective: Let $Hg \in \mathfrak{R}_H$, $f(g^{-1}H) = Hg$

Therefore, $|\mathfrak{L}_H| = |\mathfrak{R}_H|$ □

1.7.1 Exercices

Example 1.7.7. Find all left and right cosets of $\langle s, r \rangle \leq D_6$

$$1. eH = He = H$$

$$2. rH = \{r, r^3, r^5, sr, sr^3, sr^5\} = Hr$$

Since these union to the full set we are done

Example 1.7.8. Suppose that $[G : H] = 2$ and $a, b \notin H$. Show that $ab \in H$

Since $a \notin H \implies a^{-1} \notin H$

Since $a^{-1}, b \notin H \implies a^{-1}H \neq H$ and $bH \neq H$

$[G : H] = 2 \implies a^{-1}H = bH \implies$ by Theorem 1.7.3. $\forall ab \in H$

1.8 Lagrange's Theorem

Theorem 1.8.1 (Lagrange's Theorem). *If G is a finite group with $H \leq G$ then $|G| = [G : H]|H|$. In particular the order of a subgroup divides the order of a finite group*

Proof. Set $m = [G : H]$ and take $g_1, g_2, \dots, g_m \in G$ so that $g_1H, g_2H, \dots, g_mH$ are all distinct.

Also, $|g_iH| = |H| \ \forall 1 \leq i \leq m$. This is because you take all elements of H and add g_i to them.

Since $G = g_1H \cup g_2H \cup \dots \cup g_mH$ and these are all disjoint by Theorem 1.7.4

By the multiplication principle, $|G| = m|H| = [G : H]|H|$ □

Corollary 1.8.2. *If $|G| < \infty$ and $g \in G$ then $\text{ord}(g) \mid |G|$ thus $g^{|G|} = e$*

Proof. Take $H = \langle g \rangle$ and $\text{ord}(H) = \text{ord}(g)$ thus by Theorem 1.8.1 $\text{ord}(g) = \text{ord}(H) \mid |G|$

Thus by Lemma 1.3.9 $g^{|G|} = e$ □

Example 1.8.3. Suppose $|G| = 48$ and $K \leq H \leq G$ with $|K| = 8$.

What are the possible values of $|H|$.

By Theorem 1.8.1 we know $|H| \mid |G| = 48$ and $|K| = 8 \mid |H|$

Thus $|H| = 8, 16, 24, 48$

Example 1.8.4 (The Converse of Lagrange's Theorem is false).

Corollary 1.8.5. *If p is prime and $|G| = p$ then $G \cong \mathbb{Z}_p$*

Proof. Take $e \neq g \in G$ and note $\langle g \rangle \leq G$ also $|\langle g \rangle| \neq 1$

but by Theorem 1.8.1 $|\langle g \rangle| \mid p$ since p is prime $\implies |\langle g \rangle| = p \implies \langle g \rangle = G$

By Theorem 1.6.2 $G \cong \mathbb{Z}_p$ □

Corollary 1.8.6. *For all $a \in \mathbb{Z}, n \in \mathbb{N}$ s.t. $\gcd(a, n) = 1$ we have $a^{\varphi(n)} \equiv 1 \pmod{n}$*

Proof. Note $[a] \in U_n$, also $|U_n| = \varphi(n)$

$[a] \mid |U_n| = \varphi(n) \implies [a]^{\varphi(n)} = 1 \in U_n$

$\implies [a^{\varphi(n)}] = [1] \in U_n \implies a^{\varphi(n)} \equiv 1 \pmod{n}$ □

Theorem 1.8.7. *Let p be an odd prime and G a group of order $2p$ then $G \cong \mathbb{Z}_{2p}$ or $G \cong D_p$*

Proof.

Case 1: We have $g \in G$ s.t. $\text{ord}(g) = 2p \implies G = \langle g \rangle \cong \mathbb{Z}_{2p}$

Case 2: G does not have elements of order $2p$.

Assume G does not have an element of order $p \implies$ all non identity elements have order 2.

Take $e \neq a \neq b \neq e \in G$

Consider: $\{e, a, b, ab\} \leq G \implies 4 \mid 2p \implies 2 \mid p \Rightarrow \Leftarrow$

Thus, there exists a $h \in G$ s.t. $\text{ord}(h) = p$ and form $H = \langle h \rangle \leq G$

$[G : H] = \frac{|G|}{|H|} = \frac{2p}{p} = 2$. Take $g \notin H \implies G = H \sqcup gH$

Consider: g^2H

Case 2a: $g^2H = H \implies g^2 \in H$

$\implies \text{ord}(g^2) \mid p \implies \text{ord}(g^2) = 1$ thus $g^2 = e$ or $\text{ord}(g^2) = p$ thus $g^{2p} = e$

$\implies \text{ord}(g) = p$ since no element has order $2p \implies \langle g \rangle = \langle g^2 \rangle \leq H \implies g \in H \Rightarrow \Leftarrow$

$\implies \text{ord}(g) = 2$

Case 2b: $g^2H = gH \implies g^{-1}g^2 = g \in H \Rightarrow \Leftarrow$

Thus, anything not in H has order 2

$hg \in Hg \implies hg \notin H \implies \text{ord}(hg) = 2$

$e = (hg)^2 = hghg \implies g^{-1}h^{-1} = hg \implies gh^{p-1} = hg \implies hg^k = gh^{p-k}$

Cosets are a partition so

$G = \{e, h, h^2, \dots, h^{p-1}, g, gh, \dots, gh^{p-1}\} = \langle h, g \mid e = h^p = g^2, hg = gh^{p-1} \rangle = D_p$

□

1.8.1 Exercises

Example 1.8.8. Let G be a group of order pqr where p, q and r are distinct primes. If $H, K \leq G$ s.t. $|H| = pq$ and $|K| = qr$, show that $|H \cap K| = q$

Since $H \cap K \leq H$ and $H \cap K \leq K$

$$|H \cap K| \mid |H| = qp \text{ and } |H \cap K| \mid |K| = qr$$

$$\implies |H \cap K| = 1 \text{ or } q$$

Case 1: $|H \cap K| = q$

Case 2: $|H \cap K| = 1$

Consider: $HK = \{hk \mid h \in H, k \in K\}$

$|HK| = |H \times K|$ however we are over counting here if hk is the same element as another $h'k'$

$$h_1k_1 = h_2k_2 \implies h_2^{-1}h_1 = k_2k_1^{-1} \in H \cap K \text{ thus we over count by } |H \cap K|$$

$$\text{Thus } |HK| = \frac{|H||K|}{|H \cap K|} = pq^2r > pqr \Rightarrow \Leftarrow$$

$$\text{Thus } |H \cap K| = q$$

1.9 Normal Subgroups and Quotient Groups

Definition 1.9.1. Given a group G , we say N is a normal subgroup and write $N \trianglelefteq G$ if for all $g \in G, n \in N$ we have $gng^{-1} \in N \iff gN = Ng \iff gNg^{-1} = N$

In this case we can form the quotient group $G/N = \{gN \mid g \in G\}$ with $(xN)(yN) = (xy)N$

Theorem 1.9.2. Given a group G : $N \trianglelefteq G \iff G/N$ is a group

Proof. ($\Leftarrow$) Assume G/N is a group

Let $a, b, c, d \in G$ s.t. $aN = cN$ and $bN = dN$ then $abN = cdN$

$\implies ca^{-1} = j, bd^{-1} = h, abd^{-1}c^{-1} \in N$

$ahc^{-1} = ah(ja)^{-1} = aha^{-1}j^{-1} \in N \implies aha^{-1} \in N \implies N \trianglelefteq G$

($\implies$) Assume $N \trianglelefteq G$

Let $a, b, c, d \in G$ s.t. $aN = cN$ and $bN = dN \implies a \in bN \implies a = bn$ same for $c = dh$ and $dn = jd$ since $dN = Nd$

$$\begin{aligned} (aN)(cN) &= (ac)N \\ &= (bndh)N \\ &= (bdjhn)N \\ &= (bd)N \\ &= (bN)(dN) \end{aligned}$$

□

1.9.1 Exercises

Example 1.9.3. Show if $[G : H] = 2$ then $H \trianglelefteq G$. What is the isom. class of G/H ?

Assume $[G : H] = 2 \implies gH \sqcup H = G$ for $g \notin H$

Let $x \in G$

Case 1: $x \in H \implies xH = H = Hx$

Case 2: $x \in gH \implies x = gh$ for $h \in H \implies xH = ghH = gH$

Consider Hx since $g \notin H \implies Hg = gH \implies Hx = H(gh) = g(Hh) = gH = Hx$

Therefore $H \trianglelefteq G$ since $|G/H| = 2$ it is isomorphic to $\mathbb{Z}_2$

Example 1.9.4. Prove that the quotient group of a cyclic group is cyclic.

Let $G = \langle g \rangle$ be a group and $H \leq G$

Let $xH \in G/H$ s.t. $x \in G \implies x = g^k$ for $k \in \mathbb{Z}$

$\implies xH = g^kH = (gH)^k \implies G/H = \langle gH \rangle \implies$ it is cyclic

1.10 The Alternating Group

Definition 1.10.1. Two cycles are called transpositions

Definition 1.10.2. An element $\sigma \in S_n$ is called even(odd) if it can be written as the product of an even(odd) number of transpositions

Definition 1.10.3. The alternating group, A_n , is the subgroup of S_n made up of all even permutations.

$$A_n = \{\sigma \in S_n \mid \sigma \text{ is even}\}$$

Proposition 1.10.4. Every $\sigma \in S_n$ can be written as the product of transpositions.

Proof. Since σ can be written as the product disjoint cycles we only have to show that one cycles can be written as the product of transpositions.

Observe $(a_1, a_2, \dots, a_m) = (a_1, a_m)(a_1, a_{m-1}) \dots (a_1, a_2)$

A m cycle breaks into $m - 1$ transpositions, so an even length cycle is odd □

Proposition 1.10.5. If $(a_1, b_1)(a_2, b_2) \dots (a_n, b_n) = (1)$ then n is even, and thus $(1) \in A_n$

Observations:

$$(bc)(ab) = (ac)(bc) \tag{4}$$

$$(ac)(ab) = (ab)(bc) \tag{5}$$

$$(6)$$

Proof. Base Case: $n = 1 \implies (ab) \neq (1)$

$$n = 2 \implies (ab)(cd) = (1) \implies (ab) = (cd)$$

I.H. Assume for all $1 \leq k < m$ we have if $(a_1 b_1) \dots (a_k b_k) = (1)$

Assume $(a_1, b_1)(a_2, b_2) \dots (a_n, b_n) = (1)$

Set $a_m = a$ and use (1) or (2) to move the rightmost appearance of a to the left until

1. the two cycle containing a is next to its inverse

$$(a_1 b_1) \dots (ab_m) = (a_1 b_1) \dots (a_{j-1} b_{j-1})(ab)(ab)(a_{j+2}^{\cdot} b_{j+1}^{\cdot}) \dots (a_m^{\cdot} b_m^{\cdot})$$

= a product of $m - 2$ transpositions by I.H. we have $m - 2$ is even $\implies m$ is even

2. the two cycle containing a never finds an inverse

$$(a_1 b_1) \dots (ab_m) = (ab)(a_2^{\cdot} b_2^{\cdot}) \dots (a_m^{\cdot} b_m^{\cdot}) \implies a \text{ is not fixed} \implies \neq (1) \tag{6}$$

□

Proposition 1.10.6. A_n is a subgroup of order $\frac{n!}{2}$

Proof. $(1) \in A_n$ so nonempty Assume $\sigma, \mu \in A_n \implies \sigma = a_1 a_2 \dots a_r$ and $\mu = b_1 b_2 \dots b_s$ where r and s are even

$$\sigma \mu^{-1} = a_1 \dots a_r b_s \dots b_1 \quad r+s \text{ is even so } \sigma \mu^{-1} \in A_n$$

Thus, A_n is a subgroup

$$\text{Even} = \text{odd} \implies |A_n| = \frac{n!}{2} \quad \square$$

Corollary 1.10.7. $[S_n : A_n] = 2$ since $\# \text{even} = \# \text{odd} \implies A_n \trianglelefteq S_n \implies S_n / A_n \cong \mathbb{Z}_2$

Lemma 1.10.8. For $n \geq 3$, A_n is generated by 3-cycles.

Proof. Take $\sigma \in S_n$, write $\sigma = a_1 b_2 \dots a_m b_m$

$$a_j b_j = \begin{cases} (ab)(cd) = (acb)(acd) \\ (ab)(ac) = (acb) \end{cases} \implies \sigma \text{ can be written as 3-cycles} \quad \square$$

Lemma 1.10.9. Let $N \trianglelefteq A_n$ with $n \geq 3$. If N contains a 3-cycle then $N = A_n$

Proof. Note all 3-cycles can be written as products of $(12_)$. $a, b, c \in \{3, \dots, n\}$

$$\begin{aligned} (1a2) &= (12a)(12a) \\ (1ab) &= (12b)(12a)^2 \\ (2ab) &= (12b)^2(12a) \\ (abc) &= (12a)^2(12c)(12b)^2(12a) \end{aligned}$$

Take $(abc) \in N$

$$(2b)(1a)(abc)(1a)(2b) = (12c) \in N$$

Take $m \in \{3, \dots, n\}$, $(12m) = (12)(mc)(12c)^2(mc)(12) \in N \implies$ all 3-cycles are in N , but 3-cycles generate A_n so $N = A_n$ $\square$

Theorem 1.10.10. For $n \geq 5$, if $N \trianglelefteq A_n$ then $N = \{e\}$ or $N = A_n$

Proof. Take $(1) \neq \sigma \in N$ and write it as a product of disjoint cycles.

Case 1: There is a cycle of length ≥ 4

Write: $\sigma = (a_1 a_2 a_3 a_4 \dots a_m) \mu$

$(a_1 a_3 a_m) = \sigma^{-1}((a_1 a_2 a_3) \sigma (a_1 a_2 a_3)^{-1})$ since N is normal $\implies \in N \implies N = A_n$ by Lemma 1.10.8

Case 2: σ only contains 2,3-cycles

Case 2a: σ has at least two 3-cycles

$\sigma = (a_1 a_2 a_3)(a_4 a_5 a_6) \mu \implies (a_1 a_4 a_2 a_6 a_3) = \sigma^{-1}((a_1 a_2 a_4) \sigma (a_1 a_2 a_4)^{-1}) \in N$
 $\implies$ Case 1 $\implies N = A_n$

Case 2b: σ has one 3-cycle

$\sigma = (a_1 a_2 a_3) b_2 b_3 \dots b_k$

$\sigma^2 = (132) \in N \implies N = A_n$

Case 3: σ has only 2-cycles

$\sigma = (a_1 a_2)(a_3 a_4) \mu$

$(a_1 a_3)(a_2 a_4) = \sigma^{-1}((a_1 a_2 a_3) \sigma (a_1 a_2 a_3)^{-1}) \in N$

$(a_1 a_3 a_5) = (a_1 a_3)(a_2 a_4)((a_1 a_3 a_5)(a_1 a_3)(a_2 a_4)(a_1 a_3 a_5)^{-1}) \in N \implies N = A_n$

□

Definition 1.10.11. We say a group G is simple if it has no proper normal subgroup

1.10.1 Exercises

Example 1.10.12. $\mathbb{Z}_n$ is simple $\iff n$ is prime

Proof. Assume $\mathbb{Z}_n$ is simple

Since all subgroups of $\mathbb{Z}_n$ are normal $\implies$ the only subgroups of $\mathbb{Z}_n$ are $\mathbb{Z}_n$ and $\{e\}$

By Theorem 1.8.1 only $1, n \mid n \implies n$ is prime

Assume n is prime

Since all subgroups of $\mathbb{Z}_n$ are normal $\implies$ the only subgroups of $\mathbb{Z}_n$ are $\mathbb{Z}_n$ and $\{e\}$ since subgroups order have to divide n

Thus, $\mathbb{Z}_n$ is simple

□

1.11 External and Internal Direct Products

Definition 1.11.1. Given groups $(G_1, *_1), (G_2, *_2)$, their (external) direct product is the set

$$G_1 \times G_2 = \{(a, b) \mid a \in G_1, b \in G_2\}$$

with operation $*$:

$$(a, b) * (c, d) = (a *_1 c, b *_2 d)$$

Proposition 1.11.2. Suppose $a \in G_1, b \in G_2$ with $\text{ord}(a) = m$ and $\text{ord}(b) = n$ then $\text{ord}((a, b)) = \text{lcm}(m, n)$ in $G_1 \times G_2$

Proof. Set $k = \text{ord}((a, b)), l = \text{lcm}(m, n)$

$$\implies l = ms, l = nt \text{ for } s, t \in \mathbb{N}$$

$$(a, b)^l = (a^l, b^l) = ((a^m)^s, (b^n)^t) = (e_1, e_2) \implies k \mid l$$

$$(e_1, e_2) = (a, b)^k = (a^k, b^k) \implies a^k = e_1 \text{ and } b^k = e_2$$

$$\implies m \mid k \text{ and } n \mid k$$

$$\implies l \mid k \implies l = k$$

□

Corollary 1.11.3. $\text{ord}((g_1, g_2, \dots, g_n)) = \text{lcm}(\text{ord}(g_1), \dots, \text{ord}(g_n))$

Theorem 1.11.4. For $m, n \in \mathbb{N}$ we have

$$\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn} \iff \gcd(m, n) = 1$$

Proof. ($\Leftarrow$) Assume the $\gcd(m, n) = 1$, take $(1, 1) \in \mathbb{Z}_m \times \mathbb{Z}_n$

$$\text{ord}((1, 1)) = \text{lcm}(\text{ord}(1), \text{ord}(1)) = \text{lcm}(m, n) = mn$$

$$\mathbb{Z}_{mn} \cong \langle (1, 1) \rangle \leq \mathbb{Z}_m \times \mathbb{Z}_n \text{ both have } mn \text{ elements thus } \mathbb{Z}_{mn} \cong \langle (1, 1) \rangle = \mathbb{Z}_m \times \mathbb{Z}_n$$

$$(\implies) \text{ Assume } \mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n$$

Since $\mathbb{Z}_{mn}$ is cyclic, $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic

$$\implies \exists (a, b) \in \mathbb{Z}_m \times \mathbb{Z}_n \text{ s.t. } \langle (a, b) \rangle = \mathbb{Z}_m \times \mathbb{Z}_n$$

$$mn = |\mathbb{Z}_{mn}| = |\mathbb{Z}_m \times \mathbb{Z}_n| = \text{ord}((a, b)) = \text{lcm}(\text{ord}(a), \text{ord}(b)) \leq \text{lcm}(m, n) = \frac{mn}{d}$$

$$\implies mn \leq \frac{mn}{d} \implies d \leq 1 \implies d = 1$$

□

Theorem 1.11.5. If $\gcd(m,n) = 1$ then $U_{mn} \cong U_m \times U_n$

Proof. $|U_{mn}| = \phi(mn) = \phi(m)\phi(n) = |U_m||U_n| = |U_m \times U_n|$

$$\psi : U_{mn} \rightarrow U_m \times U_n, \psi(a) = (a, a)$$

$$\text{Suppose in } U_{mn} a = b \implies a \equiv b \pmod{mn} \implies mn \mid b - a \implies m \mid b - a \text{ and } n \mid b - a$$

$$\implies a \equiv b \pmod{m} \text{ and } a \equiv b \pmod{n} \implies a = b \text{ in } U_n \text{ and } U_m$$

$$\implies \psi(a) = (a, a) = (b, b) = \psi(b) \implies \text{well-defined}$$

$$\text{Homomorphism: } \psi(ab) = (ab, ab) = (a, a)(b, b) = \psi(a)\psi(b)$$

$$\text{Injective: Let } a, b \in U_{mn} \text{ s.t. } \psi(a) = \psi(b)$$

$$\implies (a, a) = (b, b) \implies a \equiv b \pmod{n} \text{ and } a \equiv b \pmod{m}$$

$$\implies a \equiv b \pmod{mn} \implies a = b \quad \square$$

Proposition 1.11.6. If G is a group, $H \leq G$, and $N \trianglelefteq G$ then $HN = \{hn \mid h \in H, n \in N\} \leq G$

Proof. Let $x, y \in HN$

$$x = hn_1, y = jn_2$$

$$xy^{-1} = hn_1(jn_2)^{-1} = hn_1n_2^{-1}j^{-1} = hnj^{-1} = hj^{-1}(jn_2j^{-1}) = hn \in HN \quad \square$$

Definition 1.11.7. Suppose $N_1, N_2 \trianglelefteq G$. We say G is an internal direct product of N_1 and N_2 if

$$1. N_1 \cap N_2 = \{e\}$$

$$2. G = N_1N_2$$

Theorem 1.11.8. If G is an internal direct product of N_1 and N_2 then $G \cong N_1 \times N_2$

Proof. Define $\phi : N_1 \times N_2 \rightarrow G$

$$\phi(n_1, n_2) = n_1n_2$$

$$\text{Homomorphism: Let } (a, b), (c, d) \in N_1 \times N_2$$

$$\phi((a, b)(c, d)) = \phi((ac, bd)) = acbd$$

$$\text{Consider: } cbc^{-1}b^{-1} \in N_2, N_1 \implies = e \implies cb = bc$$

$$\implies = abcd = \phi(a, b)\phi(c, d)$$

$$\text{Injective: } \phi(a, b) = \phi(c, d)$$

$$\implies ab = cd \implies c^{-1}a = db^{-1} \in N_2, N_2 \implies = e$$

$$\implies c = a \text{ and } b = d$$

$$\text{Surjective: Take } y \in G \implies y = ab, a \in N_1 \text{ and } b \in N_2 \quad \phi(ab) = ab = y \quad \square$$

1.12 First Isomorphism Theorem of Groups

Proposition 1.12.1. *If $\varphi : G_1 \rightarrow G_2$ is a homomorphism, $H \leq G_1$, $K \leq G_2$, and $N \trianglelefteq G_2$ then*

1. $\varphi(H) \leq G_2$

2. $\varphi(K)^{-1} \leq G_1$

3. $\varphi(N) \trianglelefteq G_1$

1. *Proof.* Let $x, y \in \varphi(H)$

$$\implies a, b \in H \text{ s.t. } x = \varphi(a), y = \varphi(b)$$

$$xy^{-1} = \varphi(a)(\varphi(b))^{-1} = \varphi(ab^{-1}) \in \varphi(H) \implies \varphi(H) \leq G_2 \quad \square$$

2. *Proof.* Let $a, b \in \varphi(K)^{-1}$

$$\implies \varphi(a), \varphi(b) \in K$$

$$\varphi(a)\varphi(b)^{-1} \in K \implies \varphi(ab^{-1}) \in K$$

$$ab^{-1} \in \varphi(K)^{-1} \implies \varphi(K)^{-1} \leq G_1 \quad \square$$

3. *Proof.* Let $n \in \varphi(N)^{-1}$ and $g \in G_1$

$$\implies \varphi(n) \in N, \varphi(g) \in G_2$$

$$\implies \varphi(g)\varphi(n)\varphi(g)^{-1} \in N \implies \varphi(gng^{-1}) \in N$$

$$\implies gng^{-1} \in \varphi(N)^{-1} \quad \square$$

Theorem 1.12.2. Suppose $\varphi : G \rightarrow G_2$ is a group homomorphism then there is a unique isomorphism $\psi : G/\ker\varphi \rightarrow \text{im}\varphi$ s.t. the diagram commutes

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & \text{im}(\varphi) \leq G_2 \\ \downarrow \pi & \nearrow \psi & \\ G/\ker\varphi & & \end{array}$$

Commutes: $\psi(g \ker\varphi) = \varphi(g)$

Proof. ψ is well-defined:

$$\begin{aligned} \text{Let } x\ker\varphi = y\ker\varphi \in G/\ker\varphi &\implies xy^{-1} \in \ker\varphi \\ \implies \varphi(xy^{-1}) = e &\implies \varphi(x) = \varphi(y) = \psi(y\ker\varphi) \end{aligned}$$

Homomorphism:

$$\psi((x\ker\varphi)(y\ker\varphi)) = \psi((xy)\ker\varphi) = \varphi(xy) = \varphi(x)\varphi(y) = \psi(x\ker\varphi)\psi(y\ker\varphi)$$

Injective:

$$\begin{aligned} \text{Let } g\ker\varphi \in \ker\psi & \\ \implies \psi(g\ker\varphi) = e & \\ \implies \varphi(g) = e &\implies g \in \ker\varphi \\ \implies g\ker\varphi = \ker\varphi &\implies \psi \text{ is Injective} \end{aligned}$$

Surjective:

$$\begin{aligned} \text{Let } y \in \text{im}\varphi &\implies \exists g \in G \text{ s.t. } \varphi(g) = y. \\ \implies \psi(g\ker\varphi) = \varphi(g) = y & \end{aligned}$$

□

1.12.1 Exercises

Example 1.12.3. Given a group G , an inner automorphism is a function $\psi_g : G \rightarrow G$ s.t. $\psi_g(x) = gxg^{-1}$

1. Show that ψ_g is an isomorphism
2. Show $\text{Inn}(G) = \{\psi_g \mid g \in G\}$ is a group.
3. Use the first isomorphism theorem to show

$$G/Z(G) \cong \text{Inn}(G)$$

Proof. 1. Injective:

Let $g \in G$ and $y \in \ker(\psi_g)$

$$\implies y = gxg^{-1} = e \text{ s.t. } x \in G$$

$$\implies x = e \implies y = e \implies \psi_g \text{ is bijective since } |G| = |G|$$

Homomorphism:

Let $a, b \in G$

$$\psi_g(ab) = gabg^{-1} = gag^{-1}gbg^{-1} = \psi_g(a)\psi_g(b)$$

2. Group

(a) Closure ✓

$$(b) \text{ Identity } \psi_e \circ \psi_g(n) = gng^{-1} = \psi_g \circ \psi_e(n) = \psi_g(n)$$

$$(c) \text{ Inverses } \psi_g \circ \psi_{g^{-1}}(n) = gg^{-1}ngg^{-1} = n = \psi_{g^{-1}} \circ \psi_g(n) = \psi_e(n)$$

(d) Associativity ✓

3. By Lemma 1.4.5 $Z(G) \leq G$

Normal:

$$gZ = Zg \implies Z(G) \trianglelefteq G$$

$$\implies G/Z(G) \text{ is a group}$$

Define: $\phi : G \rightarrow \text{Inn}(G)$, $\phi(g) = \psi_g$

$\text{Im}(\phi) = \text{Inn}(G)$ by definition

Consider: $\ker(\phi)$, $Z(G)$

$$\text{Let } x \in \ker(\phi) \implies \psi_x(g) = g \implies xgx^{-1} = g \implies xg = gx \implies x \in Z(G)$$

Let $x \in Z(G) \implies \varphi(x) = \psi_x(g) = xgx^{-1} = g \implies x \in \ker(\varphi)$
 $\implies Z(G) = \ker(\varphi)$

By Theorem [1.12.2](#), $G/Z(G) \cong \text{Inn}(G)$

□

1.13 More Isomorphism Theorems

Theorem 1.13.1 (Second Isomorphism Theorem). *Suppose $H \leq G$ and $N \trianglelefteq G$ then $H \cap N \trianglelefteq H$ and $H/H \cap N \cong HN/N$*

Proof. Suppose $h \in H$, $n \in H \cap N$

Since $N \trianglelefteq G$ and $h \in G$

$$\implies hnh^{-1} \in N$$

Since $h, n \in H \implies hnh^{-1} \in H \implies hnh^{-1} \in H \cap N$

$$\implies H \cap N \trianglelefteq H$$

Define : $\varphi : H \rightarrow HN/N$

$$h \mapsto hN$$

Homomorphism: $\varphi(xy) = (xy)N = xNyN = \varphi(x)\varphi(y)$

Surjective: Take $(hn)N \in HN/N$, for $h \in H$, $n \in N$

$(hn)N = hN$ because $h^{-1}(hn) = n \in N$

$$\varphi(h) = hN = (hn)N$$

Kernel: Let $h \in \ker(H) \trianglelefteq H \implies h \in H$

$$\varphi(h) = hN = N \implies h \in N \implies h \in H \cap N \implies \ker(H) = H \cap N$$

By Theorem 1.12.2 $H/H \cap N \cong HN/N$

□

Theorem 1.13.2 (Third Isomorphism Theorem). *If $M, N \trianglelefteq G$ where $N \leq M$ then*

$$(G/N)/(M/N) \cong G/M$$

Proof. Define: $\varphi : G/N \rightarrow G/M$

$$gN \mapsto gM$$

Well Defined: Suppose $xN = yN \in G/N \implies xy^{-1} \in N \leq M$

$$\implies \varphi(xN) = xM = yM = \varphi(yN)$$

Homomorphism: $\varphi(xNyN) = \varphi(xyN) = xyM = xMyM = \varphi(xN)\varphi(yN)$

Surjective: Suppose $gM \in G/M \implies \varphi(gN) = gM$

Kernel: Suppose $gN \in \ker(\varphi) \iff \varphi(gN) = M = gM \iff g \in M \iff gN \in M/N$

$$\ker(\varphi) = M/N$$

By Theorem 1.12.2 $(G/N)/(M/N) \cong G/M$

□

Corollary 1.13.3. $\mathbb{Z}_n / \langle m \rangle \cong \mathbb{Z}_{\gcd(m,n)}$

Proof. Set $d = \gcd(m,n)$ by Theorem 1.5.9 $\langle m \rangle = \langle d \rangle \implies \text{ord} = \frac{n}{d}$

$$\implies \cong \mathbb{Z}_{\frac{n}{d}} \cong d\mathbb{Z} / n\mathbb{Z}$$

$$\mathbb{Z}_n / \langle m \rangle \cong \mathbb{Z} / n\mathbb{Z} / d\mathbb{Z} / n\mathbb{Z} \cong \mathbb{Z} / d\mathbb{Z} \cong \mathbb{Z}_d \quad \square$$

Theorem 1.13.4 (Fourth Isomorphism Theorem). *Suppose $N \trianglelefteq G$. Every subgroup $K \leq G/N$ is the form $K = H/N$ where $N \leq H \leq G$. Further there is a bijection $\{H \leq G \mid N \leq H\} \longleftrightarrow \{K \leq G/N\}$*

Proof. Suppose $K \leq G/N$. Take $K = \{gN \mid g \in H\}$ s.t. $H = \{g \in G \mid gN \in K\}$

$$e \in H \text{ b/c } eN = N \in K \text{ b/c } K \leq G/N$$

$$\text{Suppose } x, y \in H \implies xN, yN \in K \implies xN(yN)^{-1} \in K \implies xy^{-1}N \in K \implies xy^{-1} \in H \implies H \leq G$$

Consider the projection:

$$\Pi : G \rightarrow G/N$$

$$g \mapsto gN$$

Define F by $F(H) = \Pi(H) = H/N$

Injective: Assume $F(H_1) = F(H_2)$

$$\implies H_1/N = H_2/N$$

$$\text{Suppose } h \in H_1 \iff hN \in H_1/N = H_2/N \iff h \in H_2$$

$$\implies H_1 = H_2$$

Surjective: Suppose $K \leq G/N \implies K = H/N$ s.t. $N \leq H \leq G$

$$F(H) = H/N = K \quad \square$$

1.13.1 Exercises

Example 1.13.5. If $M, N \trianglelefteq G$ s.t. $G = MN$ then $G/M \cap N \cong G/M \times G/N$

Proof. Assume $M, N \trianglelefteq G$ s.t. $G = MN$

$$\text{Let } \varphi(g) = (gM, gN)$$

Surjective: Let $(aM, bN) \in G/M \times G/N$ s.t. $a, b \in G$

$$\text{Since } G = MN \exists g \in G \text{ s.t. } g = ab \implies \varphi(g) = (gM, gN) = (aM, bN)$$

$$\text{Kernel: Let } g \in \ker(\varphi) \iff \varphi(g) = (M, N) = (gM, gN)$$

$$\iff M = gM \text{ and } N = gN \iff g \in M \cap N$$

By Theorem 1.12.2 $G/M \cap N \cong (G/M) \times (G/N)$ $\square$

1.14 Fundamental Theorem of Abelian Groups

Theorem 1.14.1 (Fundamental Theorem of Finite Abelian Groups). *If G is a finite abelian group then*

$$G \cong \mathbb{Z}_{p_1^{a_1}} \times \mathbb{Z}_{p_2^{a_2}} \times \cdots \times \mathbb{Z}_{p_n^{a_n}}$$

for (not necessarily distinct) primes p_i and $a_i \in \mathbb{N}$

Lemma 1.14.2. *Let G be an abelian group with $|G| = n$ and p a prime s.t. $p \mid n$ then G has an element of order p*

Proof. Base Case: $n = 1$

$$G = \{e\}$$

IH: Assume for all $1 \leq m < n$ the statement holds

Consider: $|G| = n$

Case 1: G has no proper subgroups. If $e \neq g \in G$

$$\{e\} \leq \langle g^{\frac{n}{p}} \rangle \leq \langle g \rangle = G$$

$$\text{Case 1a: } \{e\} = \langle g^{\frac{n}{p}} \rangle \implies e = g^{\frac{n}{p}} \implies \frac{n}{p} \text{ is a multiple of } n \implies p = 1 \implies \Leftarrow$$

$$\text{Case 1b: } \langle g^{\frac{n}{p}} \rangle = \langle g \rangle \implies p = \text{ord}(g^{\frac{n}{p}}) = \text{ord}(g) = n \implies g \text{ is order } p$$

Case 2: G has a proper subgroup $H \leq G$ s.t. $1 < |H| < n$,

also $H \trianglelefteq G$ since G is an abelian group.

Case 2a: $p \mid |H|$ then by IH we have $h \in H$ where $\text{ord}(h) = p$

Case 2b: $p \nmid |H|$ then $p \mid |G/H|$ b/c $n = |G| = |H||G/H| \Leftarrow$ Theorem 1.8.1

Apply IH to G/H $gH \in G/H$ where $\text{ord}(gH) = p$

$$\implies gH \neq H \implies g \notin H$$

$$H = (gH)^p = g^p H \implies g^p \in H. \text{ Set } |H| = m \implies (g^p)^m = e \implies (g^m)^p = e$$

Assume $\text{ord}(g^m) \neq p \implies \text{ord}(g^m) = 1 \implies g^m = e$, note $\gcd(m, p) = 1$ b/c $p \nmid m$

$$\implies mx + py = 1 \text{ for } x, y \in \mathbb{Z}$$

$$g = (g^m)^x (g^p)^y \implies g = eh \text{ for some } h \in H \implies \Leftarrow$$

$$\text{ord}(g^m) = p$$

□

Definition 1.14.3. We say G is a p -group if for all $g \in G$ $\text{ord}(g) = p^a$ for some $a \in \mathbb{Z}_{\geq 0}$

Lemma 1.14.4. G is a finite abelian p -group $\iff |G| = p^n$

Proof.

($\implies$) Assume p, q are distinct primes s.t. $p \mid |G|$ and $q \mid |G|$

By Lemma 1.14.2 find $x, y \in G$ s.t. $\text{ord}(x) = p$ and $\text{ord}(y) = q$

$\implies G$ is not a p -group

($\impliedby$) Assume $|G| = p^n$ and $g \in G$

$\langle g \rangle \leq G \implies |\langle g \rangle| \mid |G| \implies \text{ord}(g) \mid p^n \implies \text{ord}(g) = p^m$ for $0 \leq m \leq n$ □

Lemma 1.14.5. Let G be a finite abelian group of order $p_1^{r_1} p_2^{r_2} \cdots p_k^{r_k}$ for distinct primes p_i then

$$G \cong G_1 \times G_2 \times \cdots \times G_k$$

with $|G_i| = p_i^{r_i}$

Proof. Set $G_i = \{g \in G \mid \text{ord}(g) = p_i^n \text{ s.t. } n \in \mathbb{N}\}$

Subgroups: $e \in G_i, \text{ord}(e) = 1 = p_i^0$

Closure: Let $x, y \in G_i \implies \text{ord}(x) = p_i^m, \text{ord}(y) = p_i^n$

$\text{ord}(xy) = \text{lcm}(p_i^m, p_i^n) = p_i^{\max(m,n)} \implies xy \in G_i$

Inverses: Let $x \in G_i \implies \text{ord}(x) = p_i^n \implies \text{ord}(x^{-1}) = p_i^n$

$\implies G_i \leq G$

Let $g \in G_i \cap G_j$ s.t. $i \neq j \implies \text{ord}(g) = p_i^m = p_j^n \implies m, n = 0$ since these are distinct primes

$\implies g = e \implies G_i \cap G_j = \{e\}$

Let $g \in G \implies \text{ord}(g) \mid p_1^{r_1} \cdots p_k^{r_k}$

Set $a_i = p_1^{r_1} \cdots p_{i-1}^{r_{i-1}} p_{i+1}^{r_{i+1}} \cdots p_k^{r_k} = \frac{|G|}{p_i^{r_i}}$

$\text{gcd}(a_1, a_2, \dots, a_k) = 1$

$\implies$ find $b_i \in \mathbb{Z}$ s.t. $a_1 b_1 + \cdots + a_k b_k = 1$

$g = g^{a_1 b_1} \cdots g^{a_k b_k} = g_1 \cdots g_k$ where $g_i = g^{a_i b_i}$

Note: $g_i^{p_i^{r_i}} = g^{(a_i p_i^{r_i}) b_i} = g^{|G| b_i} = e \implies g_i \in G_i \implies G = G_1 G_2 \cdots G_k$

By Theorem 1.11.8, $G \cong G_1 \times \cdots \times G_k$ □

Lemma 1.14.6. *Let G be a finite abelian p -group and take $g \in G$ with maximal order, then*

$$G \cong \langle g \rangle \times K$$

for some $K \leq G$

Proof. Induction on n , where $|G| = p^n$

Base Case: Let $n = 1$, $G \cong \mathbb{Z}_p \cong \langle g \rangle \times \{e\}$

IH: Assume for $1 \leq m < n$ the statement holds

Consider $|G| = p^n$ and $g \in G$ s.t. $\text{ord}(g)$ is maximal

$$\implies \text{ord}(g) = p^k$$

If $k = n$ $\text{ord}(g) = \text{ord}(G) \implies G = \langle g \rangle \cong \langle g \rangle \times \{e\}$

If $k < n$, choose $h \in G \setminus \langle g \rangle$ of min order

$$\text{ord}(h) = p^r \leq p^k \implies \text{ord}(h^p) = p^{r-1} < \text{ord}(h)$$

$$\implies h^p \in \langle g \rangle \implies h^p = g^a \text{ for } a \geq 0$$

$$(g^a)^{p^{k-1}} = (h^p)^{p^{k-1}} = h^{p^k} = e \text{ since } p^{r-1} \mid p^k$$

$$\implies \text{ord}(g^a) = \frac{\text{ord}(g)}{\gcd(\text{ord}(g), a)} \leq p^{k-1} \implies p \mid a$$

$$\implies a = ps \implies h^p = g^{ps} \implies (g^{-s}h)^p = e \text{ set } x = g^{-s}h \neq e$$

$$\implies x^p = e \implies \text{ord}(x) = p \implies \text{ord}(h) = p$$

Take $y \in \langle g \rangle \cap \langle h \rangle \implies y = h^a, 0 \leq a \leq p-1$

If $a \neq 0$ then $\gcd(a, p) = 1$ then take $r, s \in \mathbb{Z}$ s.t.

$$ar + ps = 1. \text{ Now } y^r = (h^a)^r = h^{ar} = h^{1-ps} = h(h^p)^{-s} = h \implies h \in \langle g \rangle \Rightarrow \Leftarrow$$

$$\implies a = 0 \implies y = e \implies \langle g \rangle \cap \langle h \rangle = \{e\}$$

Take $H = \langle h \rangle$

Let $gH \in G/H$ and $|G/H| = p^{n-1}$

Set $r = \text{ord}(gH) \mid \text{ord}(g)$. Also $g^r H = (gH)^r = H \implies g^r \in H \implies g^r \in \langle g \rangle \cap \langle h \rangle$

$$\implies g^r = e \implies \text{ord}(g) \mid r \implies r = \text{ord}(g) = p^k$$

Since $\text{ord}(xH) \mid \text{ord}(x) \implies \text{ord}(gH)$ is maximal

Apply IH on $G/H \implies G/H \cong \langle gH \rangle \times J$

where by Theorem 1.13.4 $J = K/H$ s.t. $H \leq K \leq G$

Let $x \in \langle g \rangle \cap K \implies xH \in \langle gH \rangle \cap K/H \implies xH = eH \implies x \in H \implies x = e$

$$\implies \langle g \rangle \cap K = \{e\}$$

Let $x \in G \implies xH \in G/H = \langle gH \rangle (K/H)$

$$\implies xH = g^r KH \implies x \in g^r KH \subseteq \langle g \rangle K \text{ since } H \leq K$$

By Theorem 1.11.8, $G \cong \langle g \rangle \times K$

□

Fundamental Theorem of Finite Abelian Groups

Theorem 1.14.1

Proof. Assume G is a finite abelian group

By Lemma 1.14.5, $G \cong G_1 \times G_2 \times \cdots \times G_k$ where $|G_i| = p_i^{r_i}$

By Lemma 1.14.6, $G_i = \langle g_i^{(1)} \rangle \times H_i$ where $\text{ord}(g_i^{(1)}) = p_i^{s_i^{(1)}}$

By Theorem 1.6.2, $G_i \cong \mathbb{Z}_{p_i^{s_i^{(1)}}} \times H_i$

Then repeat this on H_i until $G_i \cong \mathbb{Z}_{p_i^{s_i^{(1)}}} \times \cdots \times \mathbb{Z}_{p_i^{s_i^{(m)}}}$

$$\implies G \cong \mathbb{Z}_{p_1^{a_1}} \times \mathbb{Z}_{p_2^{a_2}} \times \cdots \times \mathbb{Z}_{p_n^{a_n}}$$

□

Example 1.14.7. All abelian groups of order $8 = 2^3$.

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$$

$$\mathbb{Z}_2 \times \mathbb{Z}_4$$

$$\mathbb{Z}_8$$

1.14.1 Exercises

Example 1.14.8. All abelian groups of order $720 = 2^4 3^2 5$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$$

$$\mathbb{Z}_2 \times \mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$$

$$\mathbb{Z}_{16} \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9 \times \mathbb{Z}_5$$

$$\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_9 \times \mathbb{Z}_5$$

$$\mathbb{Z}_2 \times \mathbb{Z}_8 \times \mathbb{Z}_9 \times \mathbb{Z}_5$$

$$\mathbb{Z}_{16} \times \mathbb{Z}_9 \times \mathbb{Z}_5$$

$$\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$$

$$\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_9 \times \mathbb{Z}_5$$

Example 1.14.9. Show that $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \cdots$ is not finitely generated.

Assume it is finitely generated.

Everything has order 2 $\implies G = \mathbb{Z}_2 \times \mathbb{Z}_2 \times \cdots \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \cdots \mathbb{Z}_2 = (\mathbb{Z}_2)^n$

$G \cong (\mathbb{Z}_2)^n \times G, \{e\} \times G \trianglelefteq (\mathbb{Z}_2)^n \times G \cong G$

$G/G \cong \{e\} \cong (\mathbb{Z}_2)^n \cong (\mathbb{Z}_2)^n \times G/\{e\} \times G$

$\implies G$ has order 1. $\implies \Leftarrow$

2 Rings

2.1 Intro

Definition 2.1.1. A ring R is a set together with operations $+$ and $\cdot$ such that

- $(R, +)$ is an abelian group
- $\cdot$ is associative: For all $a, b, c \in R$ $a(bc) = (ab)c$
- There are distributive rules: For all $a, b, c \in R$ $a(b+c) = ab+ac$ and $(a+b)c = ac+bc$
- We say R is commutative: For all $a, b \in R$ $ab = ba$
- R is a "ring with an identity" if there is a $1 \in R$ s.t. $1a = a1 = a$

Proposition 2.1.2. If R has a multiplicative identity the commutativity of addition is "forced"

Proof.

$$\begin{aligned}
 a+b &= (-a) + a + a + b + b + (-b) \\
 &= (-a) + a(1+1) + b(1+1) + (-b) \\
 &= (-a) + (a+b)(1+1) + (-b) \\
 &= (-a) + a + b + a + b + (-b) \\
 &= b + a
 \end{aligned}$$

□

Example 2.1.3.

- $\mathbb{Z}_p$ (p -prime), $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$
 - Commutative
 - Has identity
 - Everything except 0 has inverse
- $\mathbb{Z}$
 - Commutative
 - Has identity
 - Only ± 1 have inverses

- $2\mathbb{Z}, 3\mathbb{Z}$, etc.
 - Commutative
 - No identity
- $\mathbb{Z}_n$
 - Commutative
 - Has Identity
 - m has an identity $\iff \gcd(m, n) = 1 \iff m \in U_n \subseteq \mathbb{Z}_n$
- $M_{n \times n}(\mathbb{R})$
 - Not Commutative (unless $n = 1$)
 - Has identity
 - A is invertible $\iff \det(A) \neq 0$

Proposition 2.1.4. *Let R be a ring with $a, b \in R$ then*

1. $0a = a0 = 0$
2. $(-a)b = a(-b) = -(ab)$
3. $(-a)(-b) = ab$
4. *If R has an identity it is unique.*

1. *Proof.* $0 = 0a + (-0a) = (0 + 0)a + (-0a) = 0a + 0a + (-0a) = 0a$

$$0 = a0 + (-a0) = a(0 + 0) + (-a0) = a0 + a0 + (-a0) = a0 \quad \square$$

2. *Proof.* $(-a)b = (-a)b + ab + (-ab) = (-a + a)b + (-ab) = 0b + (-ab) = -(ab)$

$$a(-b) = a(-b) + ab + (-ab) = a(-b + b) + (-ab) = a0 + (-ab) = -(ab) \quad \square$$

3. *Proof.* $(-a)(-b) = (-a)(-b) + (-ab) + ab$

$$= (-a)(-b) + a(-b) + ab = (-a + a)(-b) + ab = 0(-b) + ab = ab \quad \square$$

4. *Proof.* Let R have an identity $1, i \in R$

$$1 + i = i(1 + i) = i + i \implies i = 1 \quad \square$$

Let R be a ring

Definition 2.1.5. We say $a \neq 0$ is a zero division if there is $b \neq 0$ s.t.

$$ab = 0$$

Definition 2.1.6. Suppose $0 \neq 1$ in R . An element $u \in R$ is called a unit if there is $v \in R$ s.t.

$$uv = vu = 1$$

Definition 2.1.7. A commutative ring with an identity in which every non-zero element is a unit is called a field

Definition 2.1.8. Given a subset $S \subseteq R$ we say S is a subring if

- $(S, +) \leq (R, +)$
- For all $a, b \in S$, $ab \in S$

2.1.1 Exercises

Example 2.1.9. Show that it is impossible for an element of a ring to be both a zero divisor and a unit.

Proof. Let $a \in R$ and assume a is a zero divisor and a unit.

Then $a \neq 0$ and $\exists b \neq 0$ s.t. $ab = 0$

Then $\exists v$ s.t. $av = va = 1$

$b = (va)b = v(ab) = v0 = 0 \Rightarrow \Leftarrow$

□

2.2 Integral Domains